Data Analyst Capstone Project Report

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EXECUTIVE SUMMARY

- This project analyzed Stack Overflow Developer Survey data to identify current and future technology trends.
- JavaScript, SQL, TypeScript, HTML/CSS, and Python appeared among the most important programming languages.
- PostgreSQL was the leading database in both current usage and next-year interest.
- The dashboard summarized technology usage, future trends, and respondent demographics.
- The findings can support learners and decision-makers in planning technical skills for career development.

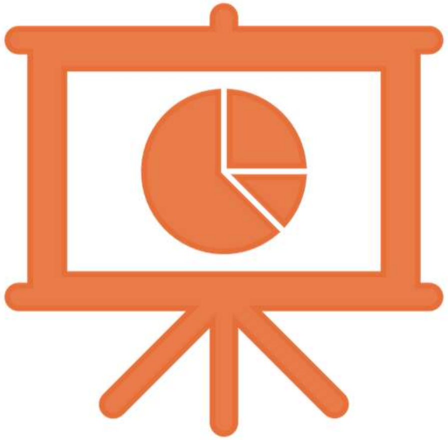
INTRODUCTION

- Purpose of the report:
  To analyze Stack Overflow Developer Survey data and identify current and future technology trends.
- Target audience:
  Learners, data analysts, and decision-makers interested in technology skills and career planning.
- Value of the analysis:
  The analysis helps highlight the most used and desired programming languages, databases, platforms, and web frameworks.
- The insights can support learning priorities, career development, and data-driven decision-making.

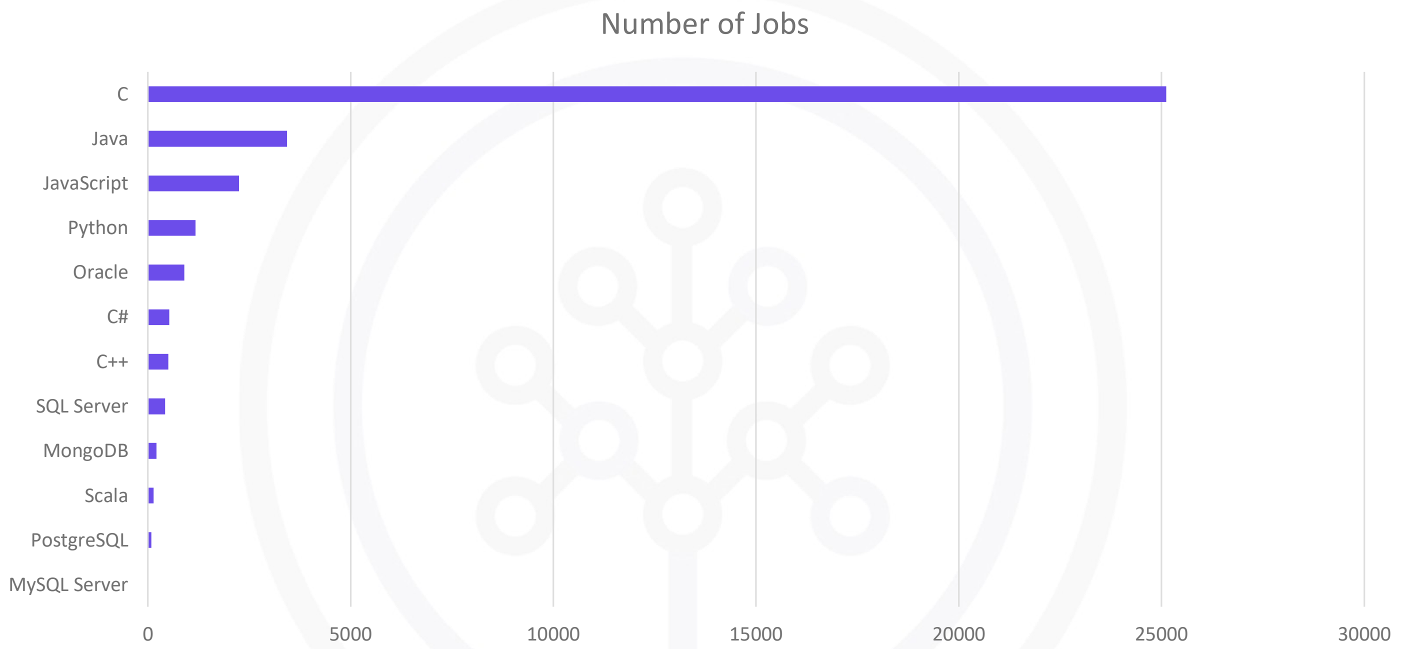
METHODOLOGY

- **Data sources**
  Stack Overflow Developer Survey dataset and job-related data collected during the capstone labs.
- **Data collection:**
  Data was collected from survey files, API outputs, and web scraping results.
- **Data wrangling:**
  The data was cleaned by checking missing values, removing duplicates, and preparing summary tables.
- **Tools used:**
  Python, Pandas, JupyterLab, Excel, and Google Looker Studio were used for analysis and visualization.

Results

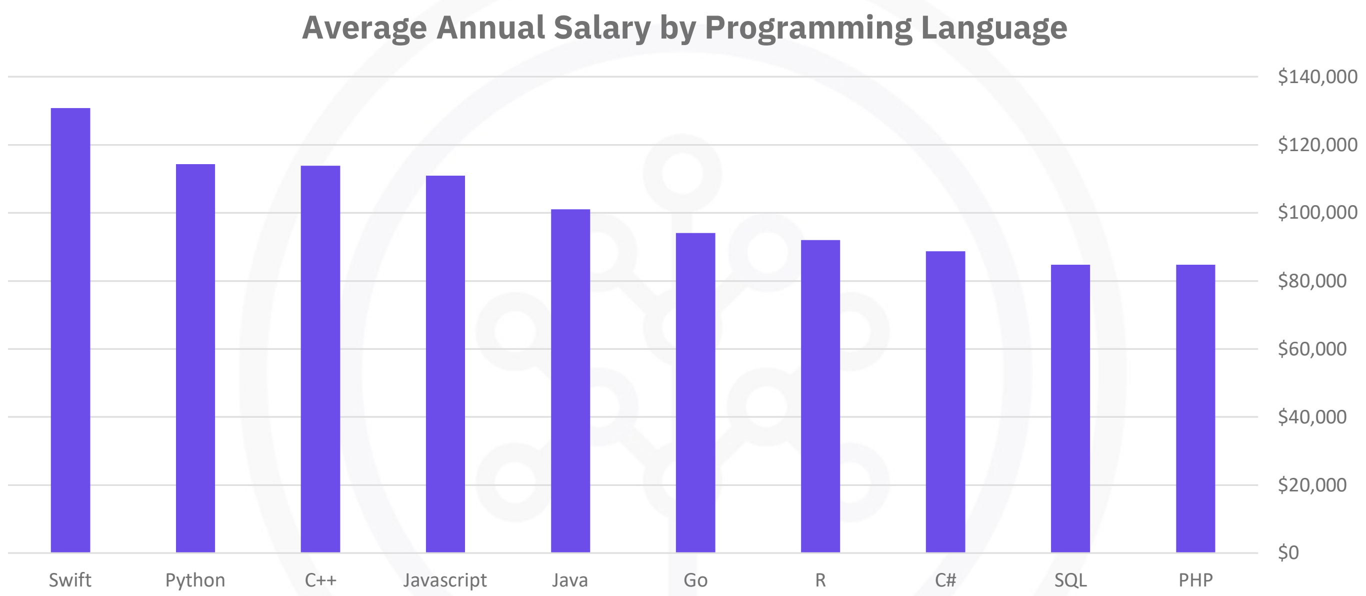


JOB POSTINGS



- **Finding:**
  C is the most in-demand technology in the job postings dataset, followed by Java and JavaScript.

POPULAR LANGUAGES

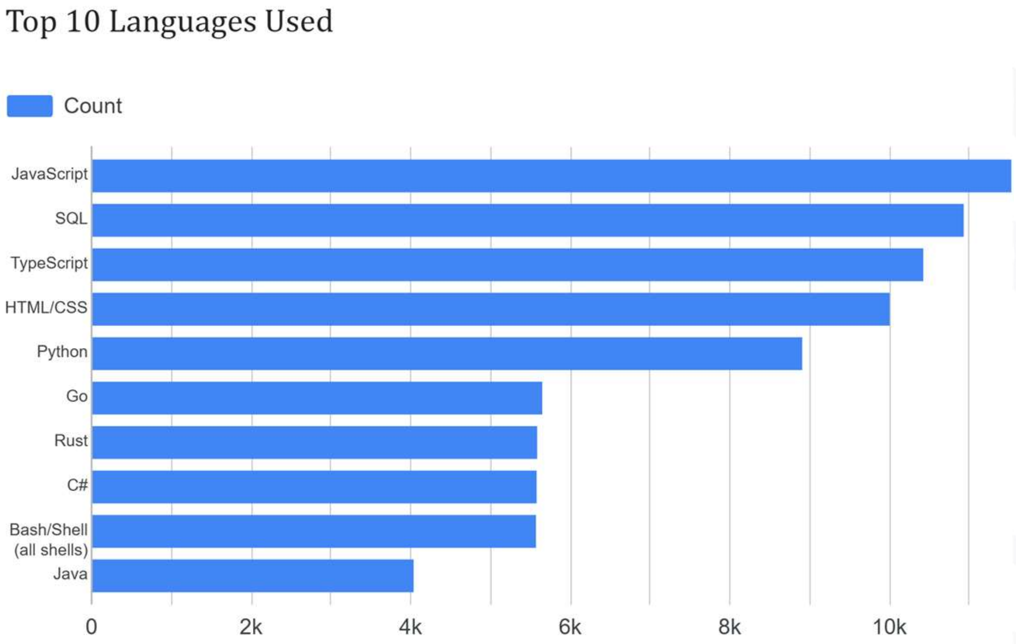


- **Finding:**
  Swift has the highest average annual salary, followed by Python and C++

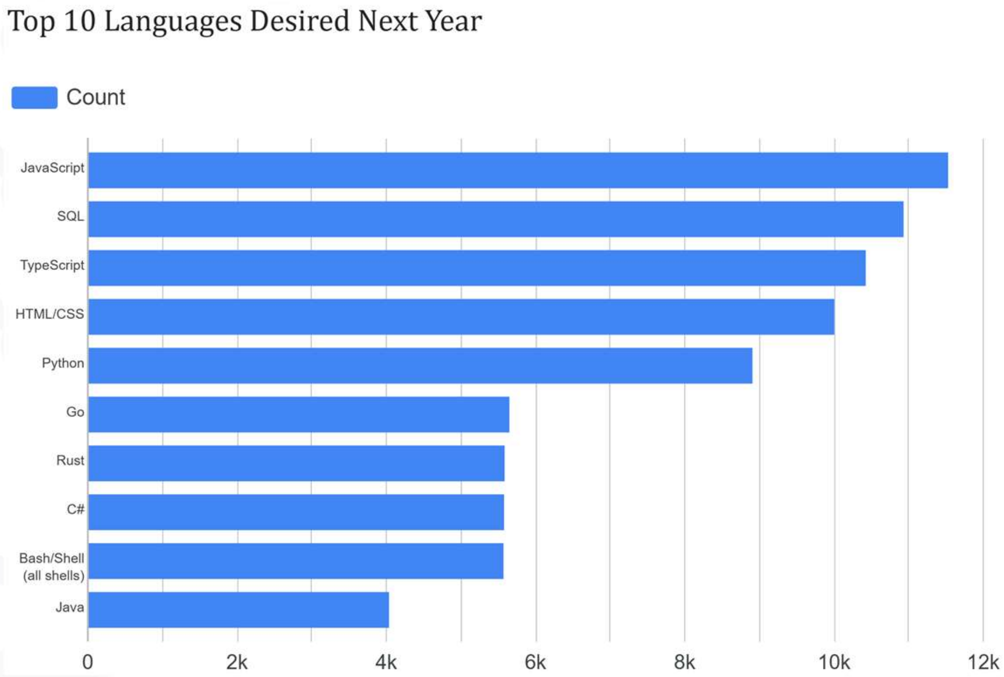
PROGRAMMING LANGUAGE TRENDS

- **Finding:**
  JavaScript, SQL, TypeScript, HTML/CSS, and Python rank highly in both current usage and next-year interest, showing continued demand for programming and web development skills.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript, SQL, TypeScript, HTML/CSS, and Python rank among the top languages in current usage.
- The same languages also appear strongly in next-year interest.
- Web development and data-related languages remain highly relevant.

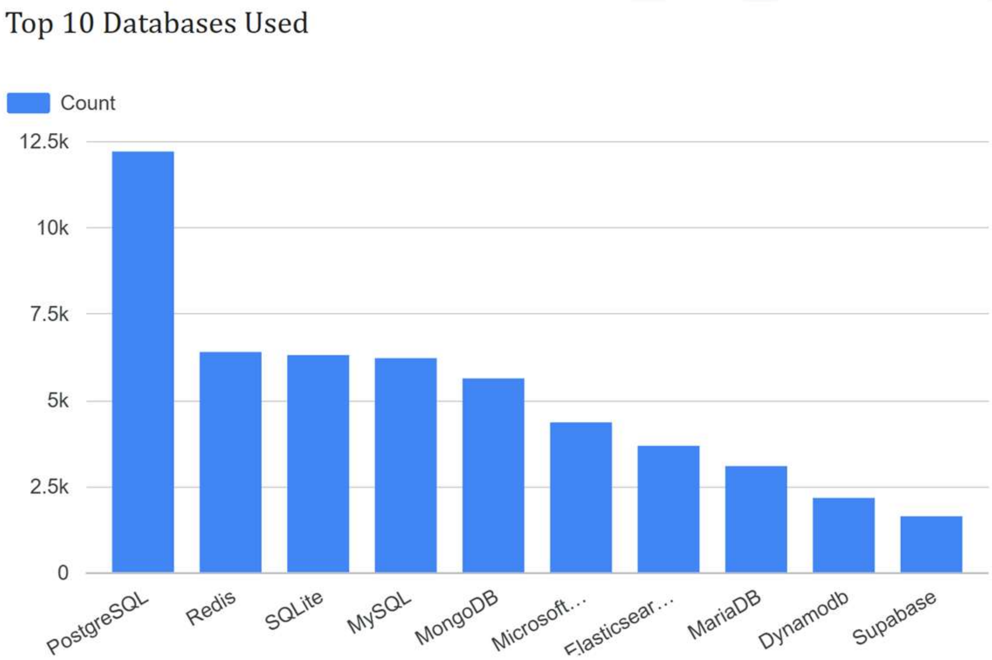
Implications

- Learners should focus on JavaScript, SQL, and Python to build strong technical skills.
- SQL and Python are especially useful for data analysis roles.
- Continued interest in these languages suggests they are valuable for future career readiness.

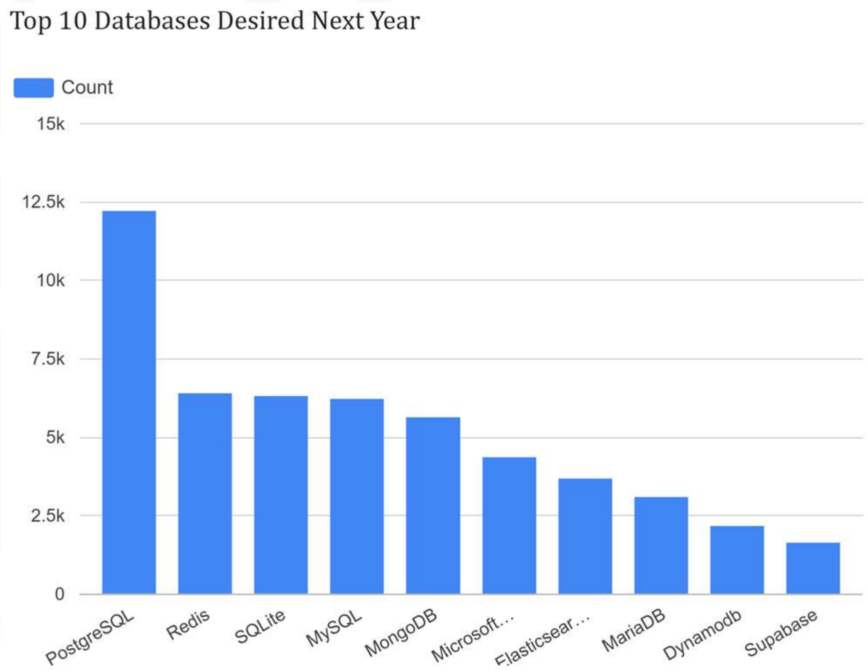
DATABASE TRENDS

- **Finding:**
  PostgreSQL ranks first in both current and future database trends, followed by Redis, SQLite, MySQL, and MongoDB.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL ranks first in both current usage and next-year interest.
- Redis, SQLite, MySQL, and MongoDB remain important database technologies.
- Current and future database preferences show similar patterns.

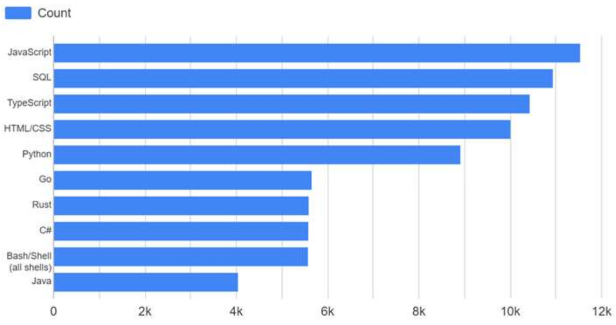
Implications

- Learners should focus on SQL-based databases such as PostgreSQL and MySQL.
- Database skills remain important for data analysis and software development roles.
- Understanding both relational and modern databases can improve career readiness.

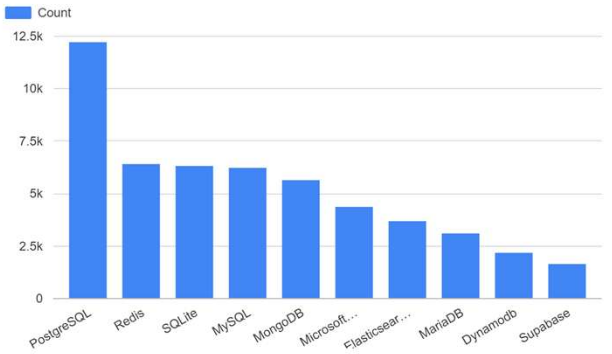
DASHBOARD

DASHBOARD TAB 1: Current Technology Usage

Top 10 Languages Used

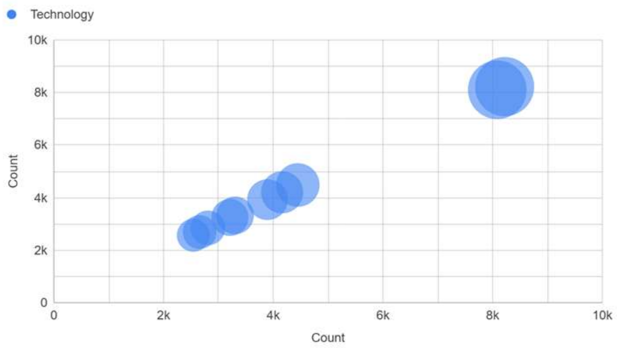


Top 10 Databases Used



Top 10 Platforms Used

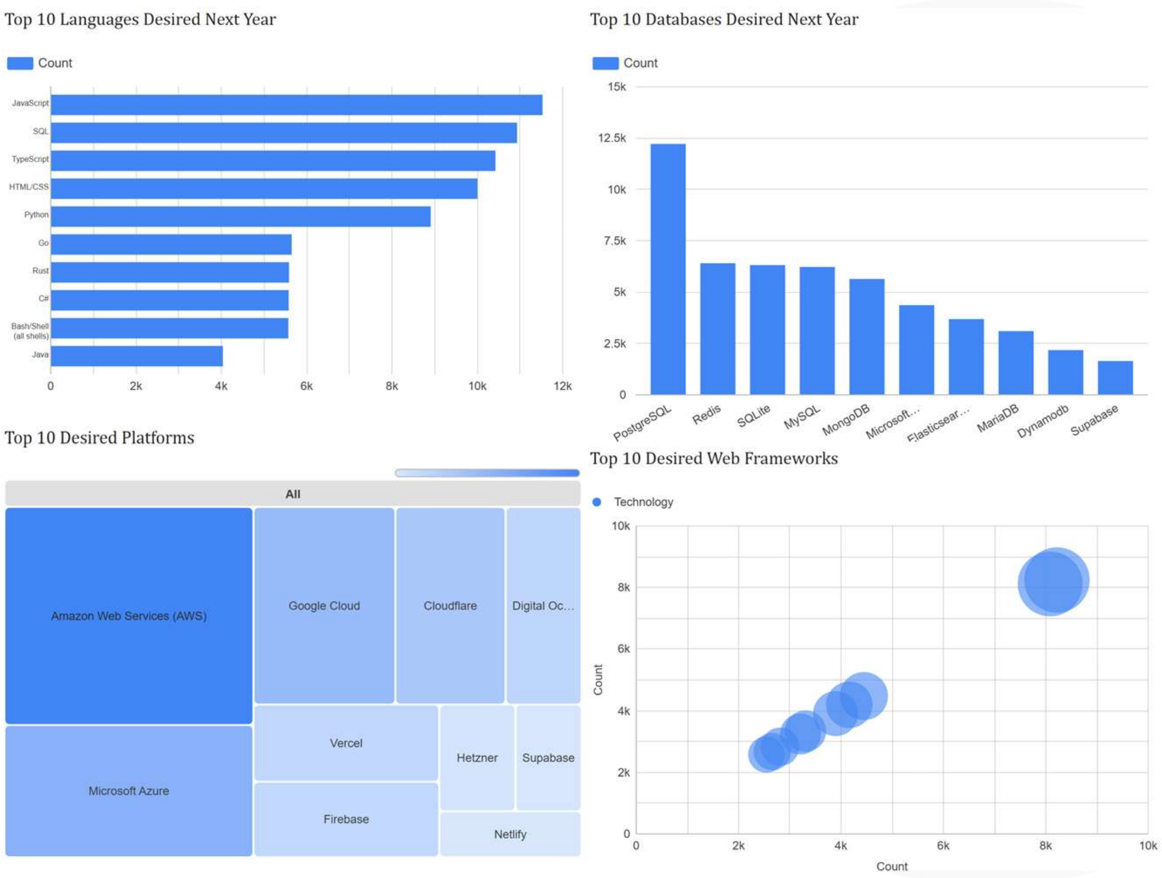
Top 10 Web Frameworks Used



Dashboard insight:

- The dashboard summarizes current technology usage across languages, databases, platforms, and web frameworks.
- JavaScript, SQL, TypeScript, HTML/CSS, and Python are the top programming languages.
- PostgreSQL is the leading database, while AWS, Microsoft Azure, and Google Cloud are key platforms.
- The results show strong current demand for programming, database, cloud, and web framework skills.

DASHBOARD TAB 2: Future Technology Trends

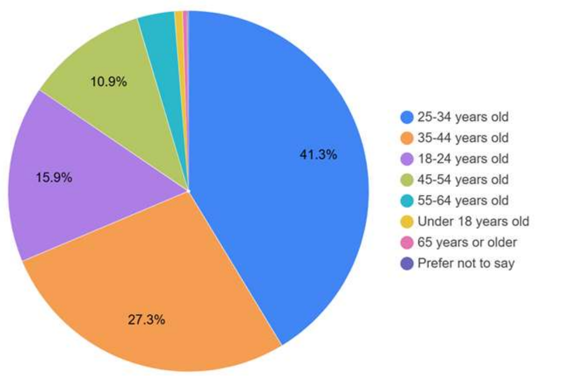


Dashboard insight:

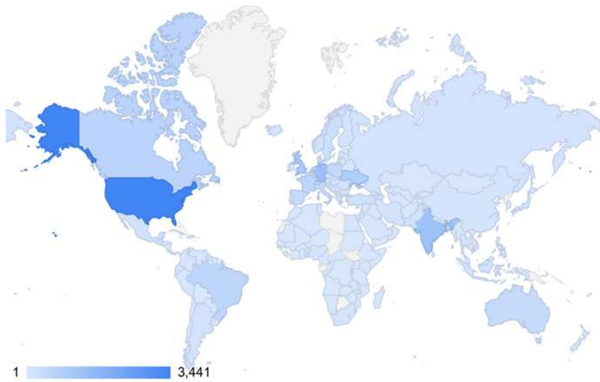
- The Future Technology Trends dashboard shows the technologies respondents want to work with next year
- JavaScript, SQL, TypeScript, HTML/CSS, and Python remain highly desired programming languages.
- PostgreSQL is the leading desired database, followed by Redis, SQLite, MySQL, and MongoDB.
- AWS, Microsoft Azure, and Google Cloud appear as important future platforms.

DASHBOARD TAB 3: Demographics

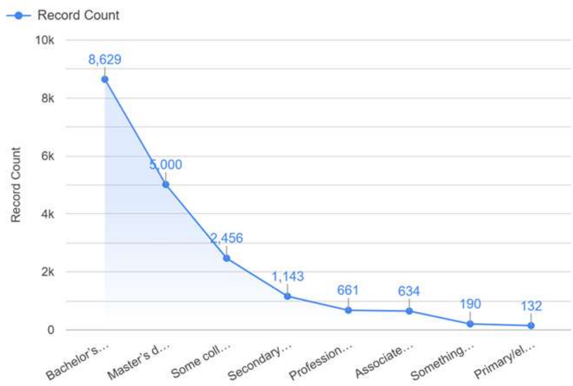
Respondent Distribution by Age



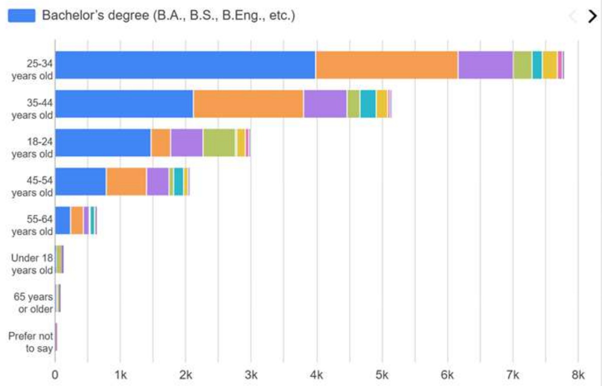
Respondent Count by Country



Respondent Distribution by Education Level



Respondent Count by Age and Education Level



Dashboard insight:

- The Demographics dashboard summarizes respondent age, country, and education level.
- Most respondents are in the 25-34 age group, followed by the 35-44 age group.
- The country map shows responses from different regions, with darker colors representing higher respondent counts.
- Education level results show that many respondents have university-level education.

DISCUSSION

- The dashboard analysis shows that programming languages, databases, cloud platforms, and web frameworks are key technology areas.
- JavaScript, SQL, TypeScript, HTML/CSS, and Python appear strongly in both current and future trends.
- PostgreSQL remains the leading database in current usage and next-year interest.
- The demographics results show that most respondents are young professionals, mainly in the 25-34 age group.
- Overall, the findings suggest that learners should focus on programming, database, cloud, and data visualization skills.

OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript, SQL, TypeScript, HTML/CSS, and Python are key programming languages.
- PostgreSQL is the leading database in both current and future trends.
- Cloud platforms such as AWS, Microsoft Azure, and Google Cloud appear important.
- Most respondents are in the 25-34 age group.

Implications

- Learners should focus on programming, database, and cloud skills.
- SQL and Python are important for data analysis careers.
- Dashboard visualizations help communicate insights clearly.
- These findings can support career planning and skill development.

CONCLUSION

- This project analyzed Stack Overflow Developer Survey data to identify technology trends.
- The findings show strong demand for programming, database, cloud, and web framework skills.
- Google Looker Studio dashboards helped present the insights clearly and effectively.
- The analysis can support learners and decision-makers in planning future technical skills.

APPENDIX

- Additional supporting charts and tables created during the analysis phase were used to support the findings presented in this report.

- These materials include dashboard screenshots, job posting summaries, and technology trend visualizations.